Lipschitz affine approximation forces exact martingale cotype

Matthew J. Colbrook

Department of Applied Mathematics and Theoretical Physics
University of Cambridge, United Kingdom
mjc249@maths.cam.ac.uk

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Abstract

We prove that the stated support-volume Lipschitz affine-approximation estimate forces martingale cotype at its exact exponent. The proof encodes finite dyadic martingales at separated spatial scales, constructs a power-convex equivalent norm and recovers the full martingale estimate.

Repository target. This manuscript concerns AIM problem 905 at revision `5d7e26e8444e`. The precise scope is stated below.

1 Main result and proof

The question asks whether the Lipschitz affine approximation estimate in [1, Question 10] determines the precise martingale cotype exponent. The Sobolev version proved in [2] has a stronger hypothesis; the argument below uses the stated Lipschitz estimate with support volume.

Let X be a real normed space of dimension $n \geq 1$, with a fixed Lebesgue measure, and let Y be a Banach space. Write B_X for its closed unit ball. For $2 \leq q < \infty$ set

$$\mathcal{D}_X(f) = \int_X \int_0^\infty \inf_{\Lambda \text{ affine}} \frac{1}{|tB_X|} \int_{x+tB_X} \|f(y) - \Lambda(y)\|^q \, dy \, \frac{dt}{t^{q+1}} \, dx.$$

Here affine maps are allowed to take values anywhere in Y.

Theorem 1.1. *Suppose there is $C < \infty$ such that every compactly supported 1-Lipschitz map $f : X \to Y$ satisfies*

$$\mathcal{D}_X(f) \leq C|\operatorname{supp} f|. \tag{1}$$

Then Y admits an equivalent uniformly convex norm whose modulus of convexity is bounded below by a positive multiple of ε^q. Consequently Y has martingale cotype q, with the exact exponent in (1).

For a compactly supported curve $F : \mathbb{R} \to Y$ define

$$\beta_F(s,u)^q = \inf_{a,c \in Y} \frac{1}{2u} \int_{s-u}^{s+u} \|F(v) - av - c\|^q \, dv, \qquad \mathcal{D}_b(F) = \int_{\mathbb{R}} \int_0^b \beta_F(s,u)^q \, \frac{du}{u^{q+1}} \, ds.$$

Lemma 1.2 (Reduction to curves)**.** *Under (1), there are $b > 0$ and $K < \infty$ such that $\mathcal{D}_b(F) \leq K$ for every 1-Lipschitz curve F supported in $[0,1]$ and satisfying $\|F\|_\infty \leq 1$.*

Proof. Choose coordinates identifying X with $\mathbb{R}^n$ and its chosen Lebesgue measure with ordinary Lebesgue measure, and numbers $b, B > 0$ such that

$$b[-1,1]^n \subseteq B_X \subseteq B[-1,1]^n.$$

Set $\kappa = (2b)^n/|B_X|$ and $L = \sum_{j=1}^n \|\ell_j\|_{X^*}$, where ℓ_j is the jth coordinate functional. Fix $R \geq \max\{1, 2B\}$ and a scalar 1-Lipschitz cutoff η taking values in $[0,1]$, equal to 1 on $[-1,1]$, and zero outside $[-2,2]$. Define

$$\chi_R(x_2,\dots,x_n) = \prod_{j=2}^n \eta(x_j/R), \qquad f_R(x) = F(x_1)\chi_R(x_2,\dots,x_n).$$

For $n = 1$ the empty product equals one. A telescoping product estimate gives $\mathrm{Lip}_X(f_R) \leq L$ and $|\operatorname{supp} f_R| \leq (4R)^{n-1}$. Scaling (1) therefore yields

$$\mathcal{D}_X(f_R) \leq CL^q(4R)^{n-1}. \tag{2}$$

If $x = (s, z')$, $z' \in [-R/2, R/2]^{n-1}$ and $0 < t \leq 1$, the cutoff equals one throughout $x + tB_X$. Restrict its local error integral to the cube $x + bt[-1,1]^n$. Averaging an affine map in the transverse coordinates produces an affine function of its first coordinate. Jensen's inequality consequently gives

$$\inf_{\Lambda} \frac{1}{|tB_X|} \int_{x+tB_X} \|f_R - \Lambda\|^q \geq \kappa\, \beta_F(s, bt)^q.$$

Integrating in s, z', t and substituting $u = bt$ shows that

$$\mathcal{D}_X(f_R) \geq \kappa R^{n-1} b^q \mathcal{D}_b(F).$$

Together with (2), this proves the lemma with $K = CL^q 4^{n-1}/(\kappa b^q)$. □

Lemma 1.3 (Separated binary scales)**.** *Suppose the conclusion of Lemma 1.2 holds. Every finite Y-valued Walsh–Paley martingale $(M_k)_{k=0}^N$ with $M_0 = 0$ and $\|M_N\|_\infty \leq 1$ satisfies*

$$\sum_{k=1}^N \mathbb{E}\|M_k - M_{k-1}\|^q \leq A, \qquad A = 36 \cdot 12^q K. \tag{3}$$

Proof. Represent its differences as

$$M_k(\epsilon_1,\dots,\epsilon_k) = M_{k-1}(\epsilon_1,\dots,\epsilon_{k-1}) + \epsilon_k a_k(\epsilon_1,\dots,\epsilon_{k-1}).$$

All values of every M_k belong to B_Y. Choose positive integers $m_1 < \cdots < m_N$, and put $h_k = 2^{-m_k}$. Require

$$\frac{3}{2}h_1 \leq b, \qquad h_{k+1} \leq \frac{1}{4}h_k, \qquad h_{k+1} \leq \frac{h_k \delta_k}{32} \quad \text{whenever } \delta_k \text{ is defined,}$$

where δ_k is the least positive norm among the finitely many values of a_k. If all its values vanish, omit the last requirement. Such a finite sequence always exists. Its gaps may depend on the martingale, but none of our final constants do.

On $[0,1)$ let $r_m(t) = (-1)^{\lfloor 2^m t \rfloor}$ and set

$$g_k(t) = M_k(r_{m_1}(t),\dots,r_{m_k}(t)), \qquad F_k(t) = \int_0^t g_k(v)\,dv \quad (0 \leq t \leq 1).$$

Extend F_k by zero off $[0,1]$. The selected binary digits are independent fair signs, so $\int_0^1 g_k = \mathbb{E}M_k = 0$. Thus these extensions are continuous, 1-Lipschitz, supported in $[0,1]$, and have supremum norm at most one. Write $F = F_N$.

For $k < N$, on each dyadic cell of length $2h_{k+1}$ the earlier selected digits are fixed and conditional averaging cancels the remaining martingale differences. Hence the integral of $g_N - g_k$ over every such cell is zero. Since its norm is at most two,

$$\|F - F_k\|_\infty \le 4h_{k+1}. \tag{4}$$

Fix a full period $I = [2jh_k, (2j+2)h_k]$ of r_{m_k} and write $c = (2j+1)h_k$. All earlier selected digits are constant on I. There is an affine function L_I such that

$$F_k(v) = L_I(v) + a_k(I)(h_k - |v - c|), \qquad v \in I.$$

Let ψ equal $+1$ on the middle half of I, and -1 on its two outer quarters. It has zero zeroth and first moments, and

$$\frac{1}{|I|}\int_I \psi(v)(h_k - |v-c|)\,dv = \frac{h_k}{4}.$$

For every affine $\Lambda : \mathbb{R} \to Y$, the triangle and Hölder inequalities imply

$$\left(\frac{1}{|I|}\int_I \|F_k - \Lambda\|^q\right)^{1/q} \ge \left\|\frac{1}{|I|}\int_I \psi(F_k - \Lambda)\right\| = \frac{h_k\|a_k(I)\|}{4}.$$

This argument allows affine approximants with values outside the span of the martingale. Using (4) and the choice of h_{k+1} gives

$$\inf_\Lambda \left(\frac{1}{|I|}\int_I \|F - \Lambda\|^q\right)^{1/q} \ge \frac{h_k\|a_k(I)\|}{8}. \tag{5}$$

For $k = N$ the tail is zero, and when $a_k(I) = 0$ the bound is immediate.

If $|s - c| \le h_k/4$ and $5h_k/4 \le u \le 3h_k/2$, the interval $[s-u, s+u]$ contains I and has length at most $3h_k$. Thus

$$\beta_F(s,u)^q \ge \frac{2}{3}\left(\frac{h_k\|a_k(I)\|}{8}\right)^q.$$

The center windows for a fixed scale are disjoint, the scale windows for distinct k are disjoint, and all lie below b. Each period therefore contributes at least

$$\frac{2}{3}\left(\frac{h_k\|a_k(I)\|}{8}\right)^q \frac{h_k}{2}\frac{h_k}{4}\left(\frac{3h_k}{2}\right)^{-q-1} = \frac{h_k\|a_k(I)\|^q}{18\cdot 12^q}$$

to $\mathcal{D}_b(F)$. The preceding binary digits have the uniform joint distribution over these periods, so

$$\sum_{I \text{ at scale } k} 2h_k\|a_k(I)\|^q = \mathbb{E}\|M_k - M_{k-1}\|^q.$$

Summing the disjoint contributions and applying Lemma 1.2 proves (3). □

Lemma 1.4 (Renorming at the same exponent)**.** *If (3) holds, then Y admits an equivalent uniformly convex norm of power type q.*

Proof. For a unit-ball-valued martingale starting at any deterministic $x \in B_Y$, subtracting x and dividing by two permits the use of (3). Its original energy is therefore at most $2^q A$. Put $H = \max\{1, 2^q A\}$ and define

$$\Phi(x) = \sup\left\{\mathbb{E}\sum_{k=1}^N \|M_k - M_{k-1}\|^q : M_0 = x,\ (M_k) \text{ a finite Walsh–Paley martingale in } B_Y\right\}.$$

The zero-step martingale is allowed. We have $0 \le \Phi \le H$ and $\Phi(-x) = \Phi(x)$. If Y is complex, the same argument gives $\Phi(\lambda x) = \Phi(x)$ whenever $|\lambda| = 1$, so the norm constructed below is also a complex norm. Join arbitrarily near-optimal martingales starting at x, y below a first fair split from $(x+y)/2$ to x and y. Their horizons can be equalized by constant steps. Taking suprema gives

$$\Phi\left(\frac{x+y}{2}\right) \ge \frac{\Phi(x)+\Phi(y)}{2} + \frac{\|x-y\|^q}{2^q}. \tag{6}$$

Let $u(x) = \Phi(0) - \Phi(x)$. Then u is even, $u(0) = 0$, and

$$0 \le u(x) \le H, \qquad u(x) \ge \|x\|^q,$$

where both lower bounds follow from (6) at $x, -x$. Moreover

$$u\left(\frac{x+y}{2}\right) \le \frac{u(x)+u(y)}{2} - \frac{\|x-y\|^q}{2^q}. \tag{7}$$

A locally bounded midpoint-convex function is continuous and convex on the open unit ball. Convexity on radial segments, first for dyadic fractions and then by continuity, gives $u(x) \le H\|x\|$ there.

Set $r = 2^{-1/q}$, $a = 1/(2H)$, and

$$K = \{x \in B_Y : u(x) \le 1/2\}.$$

The estimates just proved imply $aB_Y \subseteq K \subseteq rB_Y$. Since $r < 1$, continuity in the open ball implies that K is a closed, symmetric convex body. Its Minkowski functional p is an equivalent norm satisfying

$$ap(z) \le \|z\| \le rp(z). \tag{8}$$

For completeness, we extract a power bound for its modulus. Write $r' = (1+r)/2$ and $L = H/(1-r') = 2H/(1-r)$. The convex function u, with values in $[0, H]$ on B_Y, is L-Lipschitz on $r'B_Y$. Indeed, extend a line segment from either endpoint a distance arbitrarily close to $1-r'$ within the unit ball, and compare secant slopes using its range bound.

For $x, y \in K$ put $m = (x+y)/2$ and $\Delta = (\|x-y\|/2)^q \le r^q = 1/2$. By (7), $u(m) \le 1/2 - \Delta$. Let $\eta = \Delta/(4L)$ and $z = m/(1-\eta)$. Since $\eta \le 1/(8L)$,

$$\|z\| \le r + 2\eta \le r + \frac{1}{4L} < r', \qquad \|z - m\| \le 2\eta.$$

It follows that $u(z) \le 1/2 - \Delta + 2L\eta \le 1/2$, so $z \in K$. Consequently $p(m) \le 1 - \eta$. Whenever $p(x-y) \ge \varepsilon$, (8) implies

$$p\left(\frac{x+y}{2}\right) \le 1 - \frac{a^q}{2^{q+2}L}\varepsilon^q.$$

This proves the asserted modulus of convexity at exponent q. □

1.1 Passage to arbitrary probability spaces at the same exponent

The final implication is the classical martingale renorming theorem [3], also recalled in [2, Theorem 1]. We include its needed direction directly to keep the exponent and the probability-space quantifiers explicit.

Lemma 1.5. *Suppose a norm p satisfies $\delta_p(\varepsilon) \ge c\varepsilon^q$ for $0 \le \varepsilon \le 2$, where $q \ge 2$ and $c > 0$. Set*

$$\kappa = \min\left\{4^{-q}, \frac{c}{2 \cdot 3^q}\right\}.$$

For every finite L^q martingale on any probability space,

$$2\kappa \sum_{k=1}^{N-1} \mathbb{E}\, p(M_{k+1} - M_k)^q \le \mathbb{E}\, p(M_N)^q.$$

Proof. We first prove the homogeneous convexity inequality

$$\frac{p(x)^q + p(y)^q}{2} - p\left(\frac{x+y}{2}\right)^q \geq \kappa p(x-y)^q. \tag{9}$$

Write $A = p(x) \geq B = p(y)$, $d = p(x-y)$, and $t = (A+B)/2$. If $A - B \geq d/2$, the scalar Clarkson inequality gives

$$\frac{A^q + B^q}{2} - t^q \geq \left(\frac{A-B}{2}\right)^q \geq 4^{-q} d^q,$$

and $p((x+y)/2) \leq t$. This proves (9) in that case. For completeness, the scalar inequality follows by writing $A = t + s$, $B = t - s$: for $z \geq 1$ the function $(z+1)^q + (z-1)^q - 2z^q$ is increasing, and at $z = 1$ it is $2^q - 2 \geq 2$.

If $A - B < d/2$, the case $d = 0$ is trivial, and otherwise $B > t/2$, $A < 3t/2$. Let $u = x/A$ and $v = y/B$. The triangle inequality implies

$$p(u - v) \geq \frac{d - (A-B)}{A} \geq \frac{d}{3t}.$$

Use the modulus bound and $(x+y)/2 = (A-B)u/2 + B(u+v)/2$ to obtain

$$p\left(\frac{x+y}{2}\right) \leq t - Bc\left(\frac{d}{3t}\right)^q \leq t - \frac{c\, d^q}{2 \cdot 3^q t^{q-1}}.$$

For $0 \leq s \leq t$, $t^q - s^q \geq t^{q-1}(t-s)$. Together with $(A^q + B^q)/2 \geq t^q$, this proves (9) in the second case.

Put $F(x) = p(x)^q$. Regard Y as a real space when taking supporting functionals (use the real part of a complex norming functional). Choose a subgradient ℓ_x of F at x. Such a functional is supplied by a norming functional for x: for $x \neq 0$, take $\ell_x = qp(x)^{q-1}x^*$ with $p^*(x^*) = 1$ and $x^*(x) = p(x)$, and put $\ell_0 = 0$. The supporting-functional inequality at x, evaluated at $(x+y)/2$ and combined with (9), gives

$$F(y) \geq F(x) + \ell_x(y - x) + 2\kappa p(y-x)^q. \tag{10}$$

For a finite-valued martingale, choose ℓ_{M_k} separately at its finitely many values. Conditional expectation of the linear term in (10) is zero. Summing over k gives

$$2\kappa \sum_{k=1}^{N-1} \mathbb{E}\, p(M_{k+1} - M_k)^q \leq \mathbb{E}F(M_N) - \mathbb{E}F(M_1) \leq \mathbb{E}F(M_N).$$

Finally, finite-valued martingales suffice even on an arbitrary probability space. For each k choose a finite-valued $\mathcal{F}_k$-measurable function s_k approximating M_k in L^q within ϵ. Let $\mathcal{A}_k = \sigma(s_1, \ldots, s_k) \subseteq \mathcal{F}_k$ and set $N_k = \mathbb{E}[M_N \mid \mathcal{A}_k]$. This is a finite-valued martingale, and the tower identity gives $N_k = \mathbb{E}[M_k \mid \mathcal{A}_k]$. Since s_k is $\mathcal{A}_k$-measurable, conditional-expectation contraction yields

$$\|N_k - M_k\|_{L^q(p)} \leq 2\epsilon.$$

The finite-valued inequality therefore passes to the original finite martingale as $\epsilon \to 0$. These conditional expectations are over finite sigma-algebras and require no extra property of the Banach space. □

Proof of Theorem 1.1. Apply the first three lemmas to obtain the equivalent norm p, with $ap(z) \leq \|z\| \leq rp(z)$ and $c = a^q/(2^{q+2}L)$. Lemma 1.5 then gives

$$\left(\sum_{k=1}^{N-1} \mathbb{E}\|M_{k+1} - M_k\|^q\right)^{1/q} \leq \frac{r}{a}(2\kappa)^{-1/q}\left(\mathbb{E}\|M_N\|^q\right)^{1/q}.$$

This is the required martingale cotype inequality on every probability space, with precisely the original exponent q. □

The decisive feature is that scale gaps may depend on each finite martingale. This makes every later primitive uniformly smaller than the affine-error signal of each earlier difference, while the integral with measure du/u^{q+1} assigns the same energy weight at every chosen scale. The Bellman construction retains the exact exponent, so no passage to $q + \varepsilon$ occurs.

References

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